

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX)		**Date:24-12-24**
Time: 3HRS	**GTM-3**	**Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

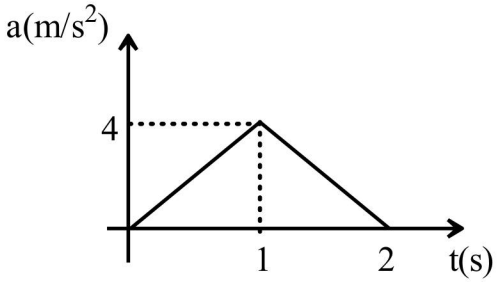
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

01. In a vernier calipers, N divisions of the main scale coincide with $N+m$ divisions of the vernier scale. What is the integral value of m for which the instrument has minimum least count?

A) 1 B) N C) $\frac{N}{10}$ D) $N/10$

02. The acceleration time graph of a particle moving in a straight line is as shown in figure. The velocity of the particle at time $t=0$ is $2\ m/s$. The velocity after $2\ s$ will be

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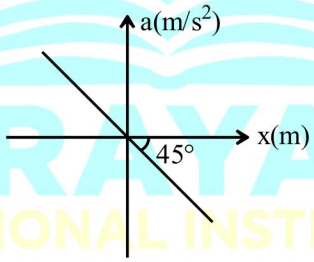


A) $6\ m/s$ B) $4\ m/s$ C) $2\ m/s$ D) $8\ m/s$

03. A tank used for filling helium balloons has a volume of $3.0 m^3$ and contains 2.0 moles of helium gas at $20.0\,^0C$. Assume that the helium behaves like an ideal gas. Then the total translational kinetic energy of the molecules of gas is___

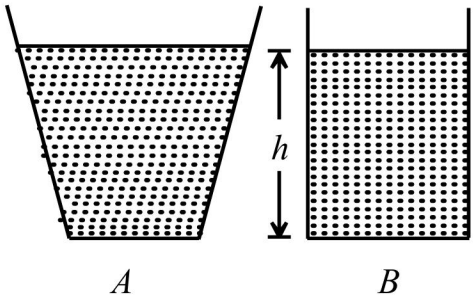
A) $7.3\times10^3 J$ B) $7.3\times10^6 J$ C) $7.3\times10^8 J$ D) $14.6\times10^4 J$

04. Acceleration – displacement graph of a particle executing SHM is as shown in given figure. The time period of its oscillation is (in s)



A) $\frac{\pi}{2}$ B) 2π C) π D) $\frac{\pi}{4}$

05. Two vessels A and B of different shapes have the same base area and are filled with water up to the same height h (see figure). The force exerted by water on the base is F_A for vessel A and F_B for vessel B. The respective weights of the water filled in vessels are W_A and W_B. Then



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A) $F_A > F_B; W_A > W_B$ B) $F_A = F_B; W_A > W_B$

C) $F_A = F_B; W_A < W_B$ D) $F_A > F_B; W_A < W_B$

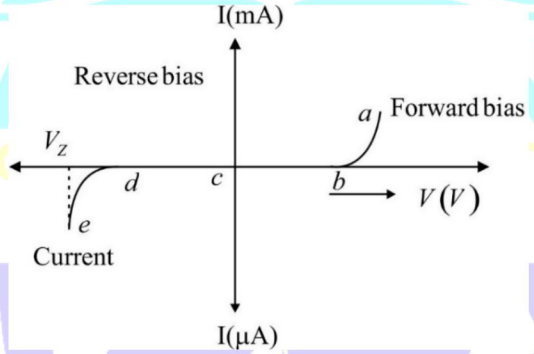
06. A homogeneous chain of length L lies on a table. The coefficient of friction between the chain and the table is μ. The maximum length which can hang over the table in equilibrium is

A) $\left(\frac{\mu}{\mu+1}\right)L$ B) $\left(\frac{\mu-1}{\mu}\right)L$ C) $\left(\frac{1-\mu}{1+\mu}\right)L$ D) $\left(\frac{2\mu}{2\mu+1}\right)L$

07. A police van moving on a highway with a speed of $30\,kmph$ fires a bullet from a gun at a thief's car which is moving away in the same direction with a speed of $190\,kmph$. If the muzzle speed (speed with respect to gun) of the bullet is $150\,ms^{-1}$, find speed of the bullet with respect to the thief's car

A) 300 kmph B) 380 kmph C) 400 kmph D) 760 kmph

08. The graph given below, represents the I – V characteristics of a Zener diode. Which part of the characteristics curve is most relevant for its operation as a voltage regulator?



A) ab B) bc C) cd D) de

09. The excess pressure inside a spherical soap bubble of radius 1 cm is balanced by a 2mm high column of oil (specific gravity = 0.8), the surface tension of the bubble is $\left(g = 9.8\ m/s^2\right)$

A) $3.92\,Nm^{-1}$ B) $0.0392\,Nm^{-1}$ C) $0.392\,Nm^{-1}$ D) $0.00392\,Nm^{-1}$

10. A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{e^2}{4\pi\varepsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and e is charge]

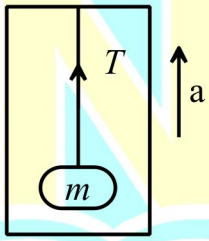
A) $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$ B) $c^2\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$ C) $\frac{1}{c^2}\left[\frac{e^2}{G4\pi\varepsilon_0}\right]^{1/2}$ D) $\frac{1}{c}G\frac{e^2}{4\pi\varepsilon_0}$

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11. A launching vehicle carrying an artificial satellite of mass **m** is set for launch on the surface of the earth of mass M and radius R. If the satellite is intended to move in a circular orbit of radius 7R, the minimum energy required to be spent by the launching vehicle on the satellite is (Gravitational constant = G)

A) $\frac{GMm}{R}$ B) $\frac{13GMm}{14R}$ C) $\frac{GMm}{7R}$ D) $\frac{GMm}{14R}$

12. A block of mass m is suspended by a light thread from an elevator. The elevator is accelerating upward with uniform acceleration a. The work done by tension on the block during t seconds is $(u=0)$



A) $\frac{m}{2}(g+a)at^2$ B) $\frac{m}{2}(g-a)at^2$ C) $\frac{m}{2}gat^2$ D) $\frac{3}{2}m(g+a)at^2$

13. In an adiabatic process, the density of a diatomic gas becomes 32 times its initial value. The final pressure of the gas is found to be n times the initial pressure. The value of n is

A) 32 B) 326 C) 128 D) $\frac{1}{32}$

14. Unit mass of a liquid of volume V_1 completely turns into a gas of volume V_2 at constant atmospheric pressure P_0 and temperature T. The latent heat of vaporization is L. Then the change in internal energy of the gas is

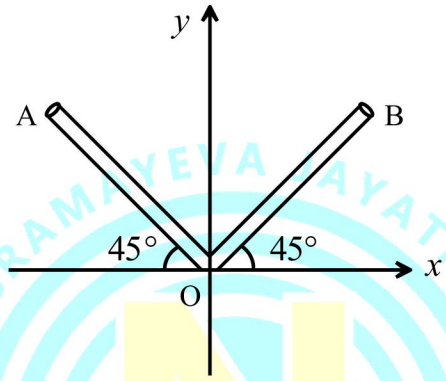
A) L B) $L+P_0(V_2-V_1)$ C) $L-P_0(V_2-V_1)$ D) Zero

15. The magnitude of torque on a particle of mass 1 kg is 2.5 Nm about the origin. If the force acting on it is 1 N, and the distance of the particle from the origin is 5m, the angle between the force and the position vector is (in radians):

A) $\frac{\pi}{6}$ B) $\frac{\pi}{3}$ C) $\frac{\pi}{8}$ D) $\frac{\pi}{4}$

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16. Two rods OA and OB of equal length and mass are lying on xy plane as shown in figure. Let I_x, I_y and I_z be the moment of inertia of both the rods about x, y and $z-$ axis respectively. Then



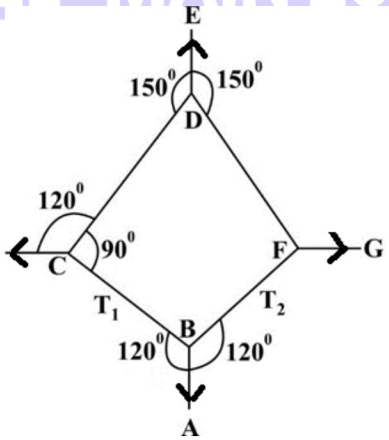
A) $I_x = I_y > I_z$ B) $I_x = I_y < I_z$ C) $I_x > I_y > I_z$ D) $I_z > I_y > I_x$

17. A composite rod made of three rods of equal length and cross-section is shown in the figure. The thermal conductivities of the materials of the rods are $K/2$, $5K$ and K respectively. The end A and end B are at constant temperatures. All heat entering the face A goes out of the end B. The effective thermal conductivity of the bar is



A) $15K/16$ B) $6K/13$ C) $5K/16$ D) $2K/13$

18. The below figure is the part of a horizontally stretched net of negligible mass. Section AB is stretched with a force of 10 N. The tensions in the sections BC and BF are



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A) 10 N, 11 N

B) 10 N, 6 N

C) 10 N, 10 N

D) Cannot be calculated due to insufficient data

19. In the figure shown, velocity of liquid which comes out is v, time of liquid to fall to ground is t and range on ground is R. If the complete setup is taken to a high mountain, match the following.

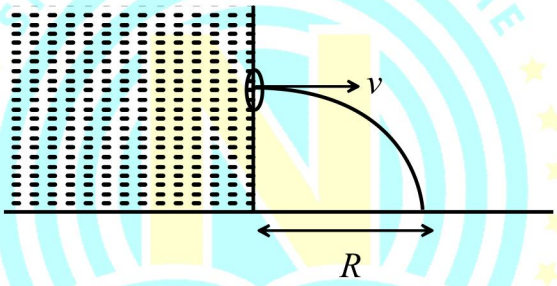


	Table-1		Table-2
a)	v	p)	will increase
b)	t	q)	will decrease
c)	R	r)	Will remain same

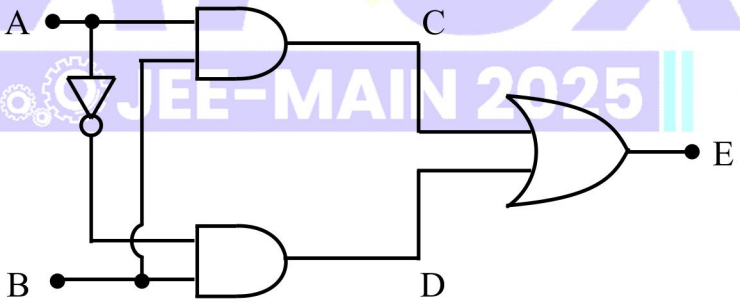
A) $a \to q, b \to p, c \to p$

B) $a \to q, b \to p, c \to r$

C) $a \to r, b \to p, c \to q$

D) $a \to r, b \to q, c \to p$

20. Truth table for the given circuit is



A)

A	B	E
0	0	1
0	1	0
1	0	1
1	1	0

B)

A	B	E
0	0	1
0	1	0
1	0	0
1	1	0

C)

A	B	E
0	0	0
0	1	1
1	0	0
1	1	1

D)

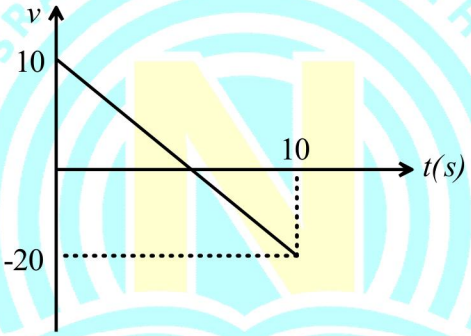
A	B	E
0	0	0
0	1	1
1	0	1
1	1	0

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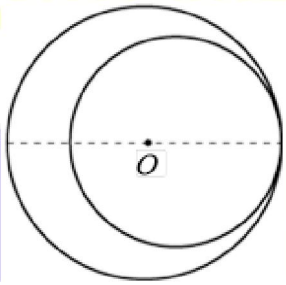
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

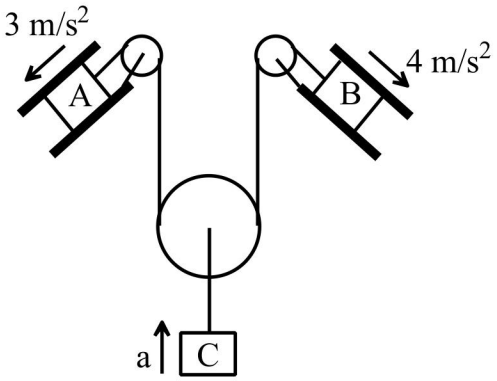
21. Velocity – time graph of a particle moving in a straight line is as shown in figure. Mass of the particle is $2\ kg$. The magnitude of work done by all the forces acting on the particle in time interval between $t = 0$ to $t = 10s$ is (in joule)



22. A circular plate of uniform thickness has a diameter of 56cm. A circular portion of diameter 42 cm is removed from one edge of the plate as shown in figure. Find the distance of centre of mass of the remaining portion from the central point O. (in cm)



23. Assuming all the surfaces to be frictionless, acceleration of the block C shown in the figure is $a\ m/s^2$ then the value of $2a$ _____



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24. If a vector A having a magnitude of 8 is added to another vector B which lies along X-axis, then the resultant of two vectors lies along Y – axis and has magnitude twice that of B. The magnitude of B is $\frac{8}{\sqrt{x}}$, then x =_____

25. A body of mass 0.5 kg travels in a straight line with velocity $v = a x^{3/2}$, where $a = 5 m^{-1/2} s^{-1}$. The work done by the net force during its displacement from x = 0 to x = 2m is (in joule)

SRAMAYEVA JAYATHE
NARAYANA
EDUCATIONAL INSTITUTIONS
APEX
JEE-MAIN 2025

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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. For the reaction, $I^- + ClO_3^- + H_2SO_4 \rightarrow Cl^- + HSO_4^- + I_2$. The **INCORRECT** statement in the balanced equation is?

A) Stoichiometric coefficient of HSO_4^- is 6

B) Iodide is oxidized

C) Sulphur is reduced.

D) H_2O is one of the products.

27. The most abundant elements in the body of a healthy human adult are Oxygen, Carbon, Hydrogen and Nitrogen. If all ^{12}C atoms are replaced by ^{14}C atoms, the weight gained by a 75 kg person is 3kg. What is the percentage by mass of carbon in the body?

A) 62 B) 24 C) 19.5 D) 2.6

28. Identify the **CORRECT** match.

(i)	XeF_2	(a)	Central atom has sp^3 hybridisation and it has bent geometry.
(ii)	N_3^-	(b)	Central atom has sp^3d^2 hybridisation and it has octahedral geometry.
(iii)	PCl_6^-	(c)	Central atom has sp hybridisation and it has linear geometry.
(iv)	ICl_2^+	(d)	Central atom has sp^3d hybridisation and it has linear geometry.

A) (i – a), (ii – b), (iii – c), (iv – d) B) (i – d), (ii – b), (iii – d), (iv – c)

C) (i – b), (ii – c), (iii – a), (iv – d) D) (i – d), (ii – c), (iii – b), (iv – a)

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29. Which reaction is spontaneous at all temperatures at standard pressure and concentration?

A) exothermic reaction with a decrease in entropy

B) exothermic reaction with an increase in entropy

C) endothermic reaction with a decrease in entropy

D) endothermic reaction with an increase in entropy

30. When 1.0 g of oxalic acid ($H_2C_2O_4$) is burned in a constant volume bomb calorimeter whose heat capacity is 8.75 kJ/K, the temperature increases by 0.312 K. The enthalpy of combustion of oxalic acid at 27°C is? (Atomic weight of C=12, H=1, O=16 & value of R=8.314 J/K/mol)

A) –245.7 kJ/mol B) –244.452 kJ/mol

C) –241.9587 kJ/mol D) –249.453 kJ/mol

31. 1 mole of N_2 and 2 moles of H_2 are allowed to react in a 1 dm^3 vessel. At equilibrium, 0.8 mole of NH_3 is formed. The concentration of H_2 at equilibrium is :

A) 0.6 mol/L B) 0.8 mol/L C) 0.2 mol/L D) 0.4 mol/L

32. For which of the gas phase reactions at 298 K, the value of $\frac{K_P}{K_C}$ is maximum and minimum respectively:

i) $N_2O_4 \rightleftharpoons 2NO_2$ ii) $2SO_2 + O_2 \rightleftharpoons 2SO_3$

iii) $X + Y \rightleftharpoons 4Z$ iv) $A + 3B \rightleftharpoons 7C$

A) i, ii B) ii, iv C) ii, iii D) iv, ii

33. In which direction will the following equilibria I and II proceed?

I : $H_2SO_4\,(aq) + NH_3\,(aq) \rightleftharpoons NH_4^+\,(aq) + HSO_4^-\,(aq)$

II : $HCO_3^-\,(aq) + SO_4^{2-}\,(aq) \rightleftharpoons HSO_4^-\,(aq) + CO_3^{2-}\,(aq)$

A) I forward & II backward B) I backward & II forward

C) Both forward D) Both backward

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34. Match the **CORRECT** atomic radius with the element:

	Element		Atomic radius (pm)
(i)	Be	(p)	74
(ii)	C	(q)	88
(iii)	O	(r)	111
(iv)	B	(s)	77
(v)	N	(t)	66

A) (i) – r, (ii) – q, (iii) – t, (iv) – s, (v) – p

B) (i) – t, (ii) – s, (iii) – r, (iv) – p, (v) – q

C) (i) – r, (ii) – s, (iii) – t, (iv) – q, (v) – p

D) (i) – t, (ii) – p, (iii) – r, (iv) – s, (v) – q

35. The properties which are not common to both groups 1 and 17 elements in the periodic table are

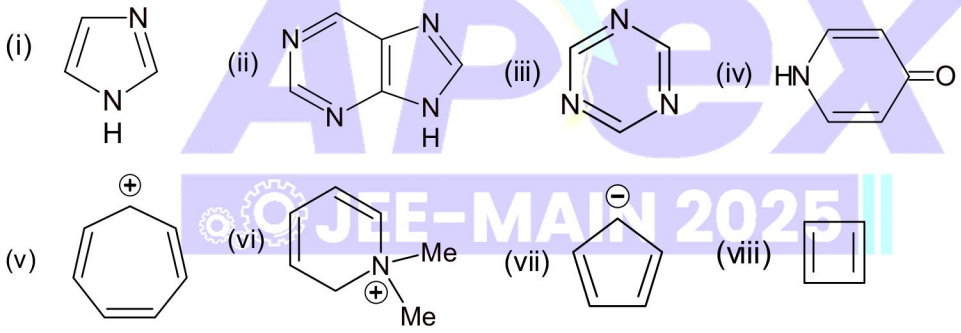
A) Electropositive character increase down the groups.

B) Reactivity decreases from top to bottom in these groups.

C) Atomic radii increase as the atomic number increase.

D) Electronegativity decreases on moving down a group.

36. Aromatic among the following compounds is/are



A) (i), (ii), (iii), (iv), (v), (vi)

B) (i), (ii), (iii), (iv), (v), (vi), (vii)

C) (i), (iii), (v), (vi), (vii), (viii)

D) (i), (ii), (iii), (iv), (v), (vii)

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37. What is the effect of pH on the solubility of $M(CN)_2$?

(neglect hydrolysis of M^{2+} ion)

A) As pH decreases, solubility decreases.

B) As pH decreases below 7 or increases above 7, solubility increases in both cases.

C) As pH decreases, solubility increases.

D) No dependence on pH of solution.

38. Stability of π-bond in following alkenes in the **increasing** order is:



(I) (II) (III) (IV)

A) I < III < IV < II

B) I < II < III < IV

C) IV < III < II < I

D) II < III < IV < I

39. Which of the following IUPAC name is **CORRECT**?

A) 4-Chloro-3-methyl cyclopentanol

B) 3-Bromo-1-amino- hexan-2-one

C) 4-chloro-3-methylcyclohexane carboxylic acid

D) 3-Bromo-1-methylhexan-1-ol

40. The sodium extract of an organic compound on acidification with acetic acid and addition of lead acetate solution gives a black precipitate. The organic compound contains

A) Nitrogen B) Halogen C) Sulphur D) Phosphorus

41.

Column-I	Column-II
(A) $CH_3CH_2COONa + NaOH \xrightarrow{CaO} ?$	(p) Wurtz reaction
(B) $CH_3MgI + C_2H_5OH \rightarrow ?$	(q) CH_4
(C) $CH_3I + Na \xrightarrow[ether]{dry} ?$	(r) C_2H_6
(D) Methane cannot be prepared as a major product	(s) Kolbes electrolysis

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A) A → r; B → q; C→ p, r; D → p,s

B) A → p; B → r; C→ p; D → p, s

C) A → p; B → r; C→ p; D → p, s

D) A → p; B → r; C→ p, q; D → s

42. Which of the following alkane has highest melting point?

A) n-pentane B) Iso pentane C) Neopentane D) n-butane

43. Which of the following statement is incorrect regarding E_2 elimination?

A) No intermediate is formed during E_2 elimination

B) It is accompanied by kinetic isotopic effect

C) The two departing (leaving) groups should be generally anti periplanar.

D) Rearrangements take place during E_2 reaction.

44. **STATEMENT-1**: Bromobenzene upon reaction with Br_2/Fe gives 1, 4-dibromobenzene as the major product

STATEMENT-2: In bromobenzene, the inductive effect of bromo group is more dominant than the mesomeric effect in directing the incoming electrophile

A) Both Statement - 1 and Statement - 2 are true and Statement - 2 is the correct explanation of Statement - 1

B) Both Statement - 1 and Statement - 2 are true but Statement - 2 is not the correct explanation of Statement - 1

C) Statement - 1 is true, Statement - 2 is false

D) Statement - 1 is false, Statement - 2 is true

45. A mixture of benzene and toluene can be separated by

A) Crystallisation B) Solubility

C) Separating funnel D) Fractional distillation

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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46. 4.3g of saturated hydrocarbon is ($C_n H_{2n+2}$) burnt completely and all CO_2 were absorbed in aq. KOH solution. A 300 ml 2N KOH solution was required. What is value of 'n' (Given: Atomic weight of C = 12, H = 1)

47. In how many of the following species, the central atom has two lone pairs of electrons?

(i) XeF_4 (ii) XeF_5^- (iii) SeO_2F_2 (iv) XeF_3^+ (v) $XeOF_4$

(vi) $ClOF_3$ (vii) ICl_4^- (viii) SCl_2 (ix) OSF_4

48. A system works in a cyclic process. It absorbs 20 calories of heat and rejects 60 J of heat during the process. The magnitude of work done (J) is [1 calorie = 4.2 J]

49. A chemist reacted 300g of 1- butene with excess Br_2 (in CCl_4) at 25^0C. He isolated 418g of 1,2- dibromobutane. The percent yield is?

(Given: Atomic weight of Br = 80, C = 12, H = 1)

50. How many of following reagents reacts with 1-butene and 2-butene separately to give same major product?

A) HBr B) Br_2 C) B_2H_6/THF, H_2O_2/OH^-

D) $Hg(OAc)_2/H_2O$, $NaBH_4/OH^-$ E) aq.H_2SO_4

F) H_2 / Ni G) O_3, H_2O/Zn H) NBS/hv

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51. The domain of the function

$f(x)=\dfrac{1}{\sqrt{[x]^2-3[x]-10}}$ is (Where $[x]$ denotes the greatest integer less than or equal to x)

A) $(-\infty,-2)\cup(5,\infty)$ B) $(-\infty,-3]\cup[6,\infty)$

C) $(-\infty,-2)\cup[6,\infty)$ D) $(-\infty,-3]\cup[5,\infty)$

52. The value of $\cos^2 10^\circ-\cos 10^\circ\cos 50^\circ+\cos^2 50^\circ$ is

A) $\dfrac{3}{4}+\cos 20^\circ$ B) $\dfrac{3}{4}$ C) $\dfrac{3}{2}(1+\cos 20^\circ)$ D) $\dfrac{3}{2}$

53. The number of elements in the set $S=\{\theta\in[0,2\pi]:\ 3\cos^4\theta-5\cos^2\theta-2\sin^6\theta+2=0\}$ is

A) 10 B) 8 C) 9 D) 12

54. A point P moves on the line $2x-3y+4=0$. If $Q(1,4)$ and $R(3,-2)$ are fixed points, then the locus of the centroid of ΔPQR is a line

A) with slope $\dfrac{3}{2}$ B) parallel to $x-$axis

C) with slope $\dfrac{2}{3}$ D) parallel to $y-$axis

55. Two sides of a parallelogram are along the lines $4x+5y=0$ and $7x+2y=0$. If the equation of one of the diagonals of the parallelogram is $11x+7y=9$, then other diagonal passes through the point

A) $(1,2)$ B) $(1,1)$ C) $(2,1)$ D) $(1,3)$

56. $\lim\limits_{t\to 0}\left(1^{\frac{1}{\sin^2 t}}+2^{\frac{1}{\sin^2 t}}+\ldots+n^{\frac{1}{\sin^2 t}}\right)^{\sin^2 t}$ is equal to

A) n^2+n B) n C) $\dfrac{n(n+1)}{2}$ D) n^2

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57. If $\lim\limits_{x\to 0}\dfrac{\alpha x e^{x}-\beta\log_e(1+x)+\gamma x^2 e^{-x}}{x\sin^2 x}=10$ (where $\alpha,\beta,\gamma\in R$), then the value of $2\alpha+3\beta-\gamma$ is ____

A) 3 B) 6 C) 18 D) 39

58. The inverse function of $f(x)=\dfrac{8^{2x}-8^{-2x}}{8^{2x}+8^{-2x}}, x\in(-1,1)$, is _____/

A) $\dfrac{1}{4}\log_e\left(\dfrac{1+x}{1-x}\right)$ B) $\dfrac{1}{4}(\log_8 e)\log_e\left(\dfrac{1-x}{1+x}\right)$

C) $\dfrac{1}{4}\log_e\left(\dfrac{1-x}{1+x}\right)$ D) $\dfrac{1}{4}(\log_8 e)\log_e\left(\dfrac{1+x}{1-x}\right)$

59. Let A,B,C be 3×3 matrices such that A is symmetric and B and C are skew-symmetric.

Consider the statements

(S1) $A^{13}B^{26}-B^{26}A^{13}$ is symmetric

(S2) $A^{26}C^{13}-C^{13}A^{26}$ is symmetric

Then

A) Only S2 is true B) Only S1 is true

C) Both S1 and S2 are false D) Both S1 and S2 are true

60. Let P be a square matrix such that $P^2=I-P.$ for $\alpha,\beta,\gamma,\delta\in N$, if $P^{\alpha}+P^{\beta}=\gamma I-29P$ and $P^{\alpha}-P^{\beta}=\delta I-13P$, then $\alpha+\beta+\gamma-\delta$ is equal to

A) 18 B) 40 C) 24 D) 22

61. If the system of equations

$x+y+z=6$

$2x+5y+\alpha z=\beta$

$x+2y+3z=14$

has infinitely many solutions, then $2\alpha+\beta$ is equal to

A) 8 B) 36 C) 44 D) 52

62. Let $[x]$ be the greatest integer $\le x$. Then the number of points in the interval $(-2,1)$, where the function $f(x)=|[x]|+\sqrt{x-[x]}$ is discontinuous is _____

A) 0 B) 1 C) 2 D) 3

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63. For $a,b>0$, let $f(x)=\begin{cases}\dfrac{\tan((a+1)x)+b\tan x}{x} & , \quad x<0\\ 3 & , \quad x=0\\ \dfrac{\sqrt{ax+b^2x^2}-\sqrt{ax}}{b\sqrt{a}\left(x\sqrt{x}\right)} & , \quad x>0\end{cases}$

be a continuous function at $x=0$. Then $\dfrac{b}{a}$ is equal to

A) 5 B) 4 C) 8 D) 6

64. Suppose a differentiable function $f(x)$ satisfies the identity

$f(x+y)=f(x)+f(y)+xy^2+x^2y$, for all real x and y. If $\lim\limits_{x\to 0}\dfrac{f(x)}{x}=1$, then $f'(3)$ is equal to

A) 10 B) 8 C) 28 D) 26

65. If $x=2\sin\theta-\sin 2\theta$ and $y=2\cos\theta-\cos 2\theta, \theta\in[0,2\pi]$, then $\dfrac{d^2y}{dx^2}$ at $\theta=\pi$ is

A) $\dfrac{3}{4}$ B) $-\dfrac{3}{8}$ C) $\dfrac{3}{8}$ D) $-\dfrac{3}{4}$

66. A point of the parabola $y^2=18x$ at which the ordinate increases at twice the rate of the abscissa is

A) $\left(\dfrac{9}{8},\dfrac{9}{2}\right)$ B) $(2,-4)$ C) $\left(\dfrac{9}{8},\dfrac{9}{4}\right)$ D) $(2,4)$

67. A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semivertical angle is $\tan^{-1}\dfrac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____

A) 8 B) 6 C) 12 D) 5

68. Let $g(x)=3f\left(\dfrac{x}{3}\right)+f(3-x)$ and $f''(x)>0$ for all $x\in(0,3)$. If g is decreasing in $(0,\alpha)$ and increasing in $(\alpha,3)$, then 8α is

A) 24 B) 0 C) 18 D) 20

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69. Let $f(x)=3\sin^4 x+10\sin^3 x+6\sin^2 x-3, x\in\left[-\frac{\pi}{6},\frac{\pi}{2}\right]$ Then, f is

A) increasing in $\left(-\frac{\pi}{6},\frac{\pi}{2}\right)$ B) decreasing in $\left(0,\frac{\pi}{2}\right)$

C) increasing in $\left(-\frac{\pi}{6},0\right)$ D) decreasing in $\left(-\frac{\pi}{6},0\right)$

70. Consider a cuboid of sides $2x$, $4x$ and $5x$ and a closed hemisphere of radius r. If the sum of their surface areas is constant k, then ratio $x:r$, for which the sum of their volumes is maximum is

A) $2:5$ B) $19:45$ C) $3:8$ D) $19:15$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. The sum of the absolute maximum and minimum values of the function $f(x)=\left|x^2-5x+6\right|-3x+2$ in the interval $[-1,3]$ is equal to

72. Let $\vec{\alpha}=4\hat{i}+3\hat{j}+5\hat{k}$ and $\vec{\beta}=\hat{i}+2\hat{j}-4\hat{k}$. Let $\vec{\beta}_1$ parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ be perpendicular to $\vec{\alpha}$. If $\vec{\beta}=\vec{\beta}_1+\vec{\beta}_2$, then the value of $5\vec{\beta}_2.\left(\hat{i}+\hat{j}+\hat{k}\right)$ is

73. Let : $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}, \vec{b}=\hat{i}-\hat{j}+2\hat{k}$ and $\vec{c}=5\hat{i}-3\hat{j}+3\hat{k}$ be three vectors. If $\vec{r}$ is a vector such that, $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{r}.\vec{a}=0$. Then $25\left|\vec{r}\right|^2$ is equal to

74. If two straight lines whose direction cosines are given by the relations $l+m-n=0, 3l^2+m^2+cnl=0$ are parallel, then the positive value of c is

75. Let $P(\alpha,\beta,\gamma)$ be the image of the point $Q(3,-3,1)$ in the line $\frac{x-0}{1}=\frac{y-3}{1}=\frac{z-1}{-1}$ and R be the point $(2,5,-1)$. If the area of the triangle PQR is λ and $\lambda^2=14K$, then K is equal to

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